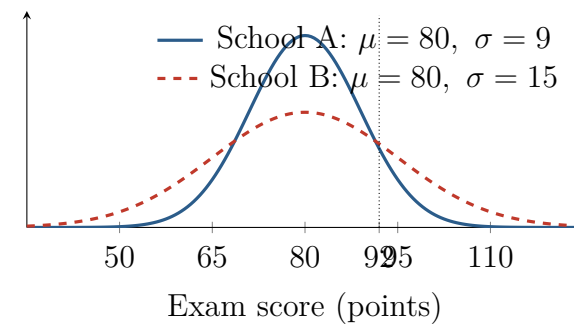


Same Score, Different Standing

Topic 5.2 · TODAY'S PROBLEM

Suppose exam scores at School A and School B are each approximately normal. Both distributions have mean 80 points, but School A has standard deviation 9 points and School B has standard deviation 15 points.

A counselor says, “A score of 92 is at about the 90th percentile at both schools because both schools have the same mean.” A data coach says, “The different spreads could put 92 at different percentiles.”



FIRST TAKE (5 min, on your sheet)

Is a score of 92 at the same percentile in both schools? Choose a claim and name one parameter behind your choice.

- DOK 1** (a) Identify which school has the more variable score distribution and state how the corresponding normal curve shows that difference.
- DOK 2** (b) For a score of 92, calculate the z -score and percentile at each school. Show the standardization and report each percentile to the nearest tenth of a percent.
- DOK 3** (c) Decide whether the counselor's same-percentile claim is warranted. Cite both z -scores and percentiles, explain why equal means do not give equal relative standing when standard deviations differ, and state at which school a 92 is 'better' if better means a higher standing relative to that school's scores. Explain what the counselor's claim would mislead a reader into believing.

Video + follow-along: [u5_lesson1-2_live.html](https://www.illustrativemathematics.org/HS/index.html) (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.

